Greyson Ekstrom

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EDUCATION

University of Georgia May 2029
B.S. Electrical & Electronics Engineering GPA: 3.84/4.00
Relevant Coursework: Circuit Analysis, Digital Logic Design, Intro to ECSE, Experiential Data Science Specialization

SKILLS

Design & Simulation: Altium Designer, KiCad, LTspice, Multisim, Fusion 360
Tools: DC Power Supply, Digital Multimeter, Soldering, 3D Printer
Hardware: Jetson Nano, Raspberry PI, Arduino
Programming: Python, MATLAB, Verilog, C++, Java

PROJECTS

Robotic Arm Dec. 2025 - Present
- Integrated Python OpenCV with YOLOv8 Nano vision processing model on Raspberry Pi which calculates and autonomously manages reverse kinematic motor controls
- Built 4-axis robotic arm controller on Arduino with CNC shield, using AccelStepper to drive stepper motors

Voltage Indicator Sep. 2026 - Present
- Engineered an offline high-voltage detector that flags >60 VDC on the FSAE EV tractive system, using an LT8315 buck converter (19-400 VDC to 12 VDC, 20 mA), resistor divider, and LM311 comparator
- Validated 12 VDC rail stability from 19-400 VDC input in LTspice, holding output within 20% and detecting the 60 VDC threshold with no false triggers

Brake System Plausibility Device Sep. 2025 - Aug. 2026
- Designed a Brake System Plausibility Device for the UGA FSAE EV that shuts the car down within 450 ms when braking and tractive-system current occur together, as FSAE rules require
- Simulated, iterated, and optimized component selection using LTspice and Altium Designer reducing switching time by 20%

ENGINEERING EXPERIENCE

FSAE EV Powertrain Controls Lead, UGA Motorsports Aug. 2026 - Present
- Led a team to integrate the vehicle's sensor circuitry, selecting and validating multiple components to meet FSAE EV rules and pass technical inspection
- Engineered low voltage sensor processing circuitry using RC filters to steady power and reduce noise from analog sensor output voltages leading to cleaner circuit response to system readings

FSAE EV Electronic Circuit Designer, UGA Motorsports Sep. 2025 - May 2026
- Collaborated among mechanical and electrical teams to develop a printed circuit board using Altium Designer which fulfilled mechanical integration requirements and competition design regulations
- Optimized circuit design and component selection using LTspice simulation to reduce net power consumption while maintaining the critical function of the circuitry

IEEE Hardware Competition Electrical Team, UGA Robotics Sep. 2026 - Present
- Co-designed the electrical system for an autonomous competition robot on an 8-person IEEE robotics team, using a Raspberry Pi as the main computer to navigate a track, avoid obstacles, and score points without human input.
- Selected and integrated motors with encoders, an IMU, servos, motor controllers, a camera, power supply, and circuit breaker under a $600 budget, using cost-versus-points analysis of game objects to optimize the design's cost-to-score ratio.

RoboDawg Electrical Team, UGA Robotics Aug. 2025 - May 2026
- Implemented software and electrical communication between motor controllers and onboard Jetson Nano using Linux
- Setup, tested, and troubleshooted persistent software errors and faulty hardware systems during initialization

Software Lead, Flying Legion FIRST Robotics Aug. 2023 - May 2025
- Led a team of 8 developers to design and optimize Java robotic control software running PhotonVision vision processing, delivering the system within a 6-week deadline and surpassing targeted performance markers
- Implemented subsystem control software communicating with brushless motors, encoder feedback loops, PID controls, and servos using Git multibranch workflow to streamline integration and improve subsystem functionality
- Developed autonomous alignment software to optimize robot task completion increasing speed and precision by up to 60%

OTHER EXPERIENCE

Summer Camp Coach, Warner Robins BEST May 2026 - Aug. 2026
- Mentored groups of campers, fostering teamwork, leadership, and positive social development in an inclusive environment

AWARDS

University of Georgia Dean's List Dec. 2025, May 2026